# Faraday's law, Displacement current, Maxwell’s equations

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# Faraday's law

- A time-varying magnetic field produces an *electromotive force* (emf) that may establish a current in a suitable closed circuit.
- An electromotive force is merely a voltage that arises from conductors moving in a magnetic field or from changing magnetic fields
- Faraday's law is customarily stated as

$$\text{emf} = -\frac{d\Phi}{dt}\ \text{V}$$

$$\text{emf} = -N\frac{d\Phi}{dt}$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L}$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\frac{d}{dt}\int_s \mathbf{B} \cdot d\mathbf{S}$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_s \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Applying Stokes' theorem

$$\int_s (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = -\int_s \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

This is one of Maxwell's four equations as written in differential, or point, form

# Review: Maxwell's four equations as they apply to static electric fields and steady magnetic fields

| | Point form | Integral form |
|---|---|---|
| 1 | $\nabla \cdot \mathbf{D} = \rho_v$ | $\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv$ |
| 2 | $\nabla \times \mathbf{E} = 0$ | $\oint \mathbf{E} \cdot d\mathbf{L} = 0$ |
| 3 | $\nabla \times \mathbf{H} = \mathbf{J}$ | $\oint \mathbf{H} \cdot d\mathbf{L} = I = \int_S \mathbf{J} \cdot d\mathbf{S}$ |
| 4 | $\nabla \cdot \mathbf{B} = 0$ | $\oint \mathbf{B} \cdot d\mathbf{S} = 0$ |

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_s \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Note: For time varying fields, 2 and 3 are different

# DISPLACEMENT CURRENT

- Modifying Ampere's Law for Time-Varying Fields $\nabla \times \mathbf{H} = \mathbf{J}$
- Taking the divergence of each side $\nabla \cdot \nabla \times \mathbf{H} \equiv 0 = \nabla \cdot \mathbf{J}$
- WKT, the equation of current continuity $\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$
- Let $0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{G}$ such that $\nabla \cdot \mathbf{G} = \frac{\partial \rho_v}{\partial t}$ Displacement current density G = $J_d$

Replacing $\rho_v$ with $\nabla \cdot \mathbf{D}$, $\nabla \cdot \mathbf{G} = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{D}) = \nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Maxwell's equation

Note: $\mathbf{J} = \sigma \mathbf{E}$ $\mathbf{J} = \rho_v \mathbf{v}$ $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$

# Maxwell's equation for time varying fields

| Point form | | Integral form |
|---|---|---|
| $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ | Surface integral and Stoke's theorem | $\oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$ |
| $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ | | $\oint \mathbf{H} \cdot d\mathbf{L} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$ |
| $\nabla \cdot \mathbf{D} = \rho_v$ | Volume integral and Divergence theorem | $\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \rho_v dv$ |
| $\nabla \cdot \mathbf{B} = 0$ | | $\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$ |

**D9.1.** Within a certain region, $\epsilon = 10^{-11}$ F/m and $\mu = 10^{-5}$ H/m. If $B_x = 2 \times 10^{-4} \cos 10^5 t \sin 10^{-3} y$ T: (*a*) use $\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t}$ to find **E**; (*b*) find the total magnetic flux passing through the surface $x = 0, 0 < y < 40$ m, $0 < z < 2$ m, at $t = 1$ μs; (*c*) find the value of the closed line integral of **E** around the perimeter of the given surface.

**Ans.** (*a*) $-20\,000 \sin 10^5 t \cos 10^{-3} y \mathbf{a}_z$ V/m;

(*b*) 0.318 mWb;

(*c*) −3.19 V

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

# Maxwell's equations phasor form

$$\mathbf{E} = \mathrm{Re}\left\{\tilde{\mathbf{E}} e^{j\omega t}\right\}$$

$$\mathbf{B} = \mathrm{Re}\left\{\tilde{\mathbf{B}} e^{j\omega t}\right\}$$

$$\nabla \times \left[\mathrm{Re}\left\{\tilde{\mathbf{E}} e^{j\omega t}\right\}\right] = -\frac{\partial}{\partial t}\left[\mathrm{Re}\left\{\tilde{\mathbf{B}} e^{j\omega t}\right\}\right]$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\mathrm{Re}\{[\nabla \times \tilde{\mathbf{E}}] e^{j\omega t}\} = -\mathrm{Re}\left\{\tilde{\mathbf{B}} \frac{\partial}{\partial t} e^{j\omega t}\right\}$$

$$= -\mathrm{Re}\left\{\tilde{\mathbf{B}}\, j\omega e^{j\omega t}\right\}$$

$$= \mathrm{Re}\left\{[-j\omega \tilde{\mathbf{B}}] e^{j\omega t}\right\}$$

- Similarly

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

- No change in Gauss's law

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega \tilde{\mathbf{D}}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega \tilde{\mathbf{B}}$$


Thank
you